EXP.NO: 1

Web Page Creation using HTML

AIM:

To create a web page that embeds a map with hotspots, frames, and links. The web page should be designed using HTML, incorporating embedded, external, and inline CSS for styling. This exercise aims to enhance understanding of integrating multimedia elements like maps, creating interactive hotspots, and organizing content using frames while applying different CSS techniques for styling.

ALGORITHM:

1)**Create the HTML Structure:**

- Set up the basic HTML document using <!DOCTYPE html>, <html>, <head>, and <body> tags.
- In the <head> section, link the external CSS file using the <link> tag, define the title, and include necessary metadata.
- In the <body>, create the required elements:

Frames:
Use the <frameset> tag to create a frame layout. (Since <frameset> is obsolete in modern HTML, <iframe> is recommended.)

Map:
Use the <img> tag to display the map image.

Hotspots:
Create clickable regions on the image using the <map> and <area> tags.

Links:
Add hyperlinks using the <a> tag to navigate to different sections of the page or to external websites.

2) **Create CSS Styles:**

- **External**
  Define styles such as colors, fonts, spacing, and layout in a separate CSS file to maintain a consistent design across the webpage.

- **Inline**
  Apply styles directly within HTML elements using the style attribute (for example, inside or <p> tags) when specific styling is required.
- **Embedded**
  Use the <style> tag inside the <head> section to define styles for particular elements such as maps, frames, and text.

3) **Map with Hotspots:**

- Use the <map> tag to create an image map.
- Define clickable areas using the <area> tag and specify the coordinates with the coords="x,y,x2,y2" attribute to mark the hotspot regions on the image.

4) **Frames:**

- Use the <iframe> tag to embed external content such as maps or webpages.
- Important attributes include:
  - src – URL of the page to display
  - width – Width of the frame
  - height – Height of the frame

5) **Add Links:**

- Inside the <body>, use the <a> tag to create navigation links that direct users to other sections of the webpage or external sites.

6) Test the Page:

- Test the webpage to ensure that frames, hotspots, and links function correctly.
- Verify that the hotspots on the map are clickable and that the linked content loads properly.
- Confirm that all styles are applied correctly using inline, embedded, and external CSS.

CODE:

Index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>University Result System | All-in-One</title>

<style>
    :root {
        --sidebar-color: #2c3e50;
        --accent: #3498db;
        --success: #27ae60;
    }

    body {
        font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
        margin: 0;
        display: flex; /* Flexbox simulates the 'Frames' structure */
        height: 100vh;
        background-color: #eef2f3;
    }

    /* Side Frame Styling */
    nav {
        width: 240px;
        background-color: var(--sidebar-color);
        color: white;
        padding-top: 20px;
        box-shadow: 2px 0 10px rgba(0,0,0,0.1);
    }
```

```
nav h2 { text-align: center; font-size: 1.1rem; border-bottom: 1px solid #444; padding-bottom: 10px; }

.nav-link {
    display: block;
    padding: 15px 25px;
    color: #bdc3c7;
    text-decoration: none;
    cursor: pointer;
    transition: 0.3s;
}

.nav-link:hover {
    background-color: #34495e;
    color: white;
    padding-left: 35px;
}

/* Main Frame Styling */
main {
    flex: 1;
    padding: 30px;
    overflow-y: auto;
}

.container {
    background: white;
    max-width: 800px;
```

```
    margin: 0 auto;
    padding: 30px;
    border-radius: 12px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.05);
}

/* Result Specific Styling */
.search-area { display: flex; gap: 10px; margin-bottom: 25px; }
input[type="text"] { flex: 1; padding: 12px; border: 2px solid #ddd; border-radius: 6px; }

button {
    padding: 12px 24px;
    background: var(--accent);
    color: white;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-weight: bold;
}

table { width: 100%; border-collapse: collapse; margin-top: 15px; }
th, td { padding: 12px; text-align: left; border-bottom: 1px solid #eee; }
th { background: #f8f9fa; color: #555; }

.error { color: #e74c3c; margin-top: 10px; font-weight: bold; }

/* Map and Hotspot Styling */
.map-box {
```

```
            border: 2px solid #eee;
            border-radius: 8px;
            overflow: hidden;
            margin-top: 20px;
        }
    </style>
</head>
<body>

    <nav>
        <h2>RESULT-PORTAL</h2>
        <a class="nav-link" onclick="toggleView('search')">Search Results</a>
        <a class="nav-link" onclick="toggleView('campus')">Campus Map</a>
        <a class="nav-link" onclick="toggleView('support')">Support Hotspots</a>
        <hr style="border: 0.5px solid #444; margin: 10px 20px;">
        <a href="https://www.rajalakshmi.org/" target="_blank" class="nav-link">Official Website ↗</a>
    </nav>

    <main>
        <div class="container">

            <section id="search-view">
                <h1 style="color: var(--sidebar-color); border-bottom: 3px solid var(--accent); padding-bottom: 10px;">
                        Examination 2025-2026
                </h1>

                <div class="search-area">
```

```
            <input type="text" id="rollNo" placeholder="Enter Roll Number (e.g. 101, 102, 103)">
            <button onclick="findResult()">Find Result</button>
          </div>

          <div id="errorMsg" class="error"></div>

          <div id="resultCard" style="display:none; animation: fadeIn 0.5s;">
            <div style="background: #f8f9fa; padding: 15px; border-radius: 8px; display: flex; justify-content: space-between;">
              <span><strong>Name:</strong> <span id="sName"></span></span>
              <span><strong>GPA:</strong> <span id="sGPA" style="color: var(--success); font-weight: bold;"></span></span>
            </div>
            <table>
              <thead>
                <tr><th>Subject</th><th>Score</th><th>Grade</th></tr>
              </thead>
              <tbody id="resultTableBody"></tbody>
            </table>
          </div>
        </section>

        <section id="campus-view" style="display:none;">
          <h1 style="color: var(--sidebar-color);">Campus Location</h1>
          <p>Find our physical examination center below:</p>
          <div class="map-box">
            <iframe
```

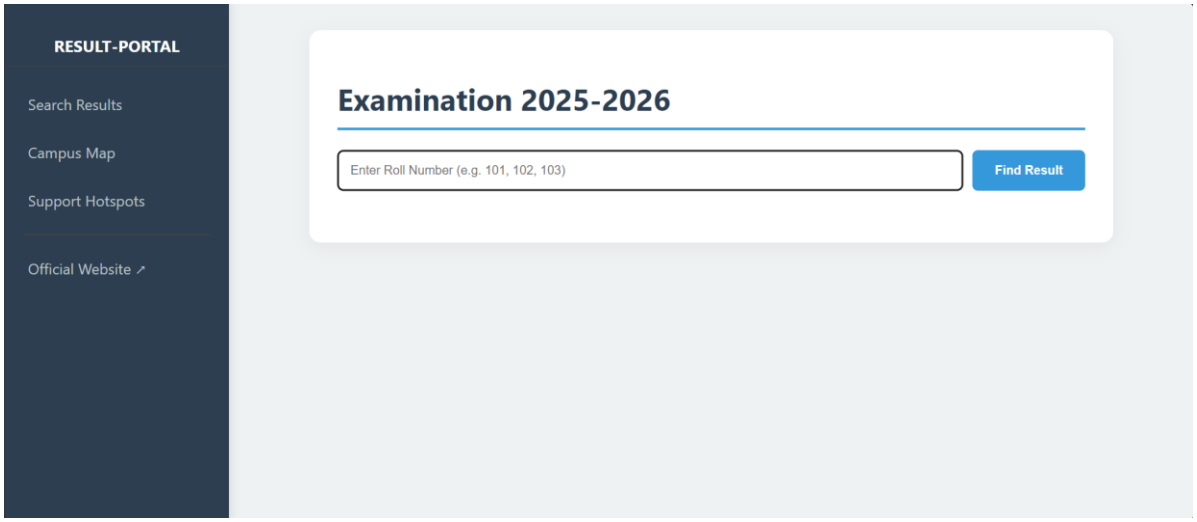
src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d3887.4258203018458!2d80.00089457507708!3d13.008533387310248!2m3!1f0!2f0!3f0!3m2!1i1024!2i768!4f13.1!3m3!1m2!1s0x3a528c9ebac84723%3A0x18e2bf88dfefa3ed!2sRajalakshmi%20Engineering%20College!5e0!3m2!1sen!2sin!4v1770375025376!5m2!1sen!2sin" width="600" height="450" style="border:0;" allowfullscreen="" loading="lazy" referrerpolicy="no-referrer-when-downgrade"></iframe>"

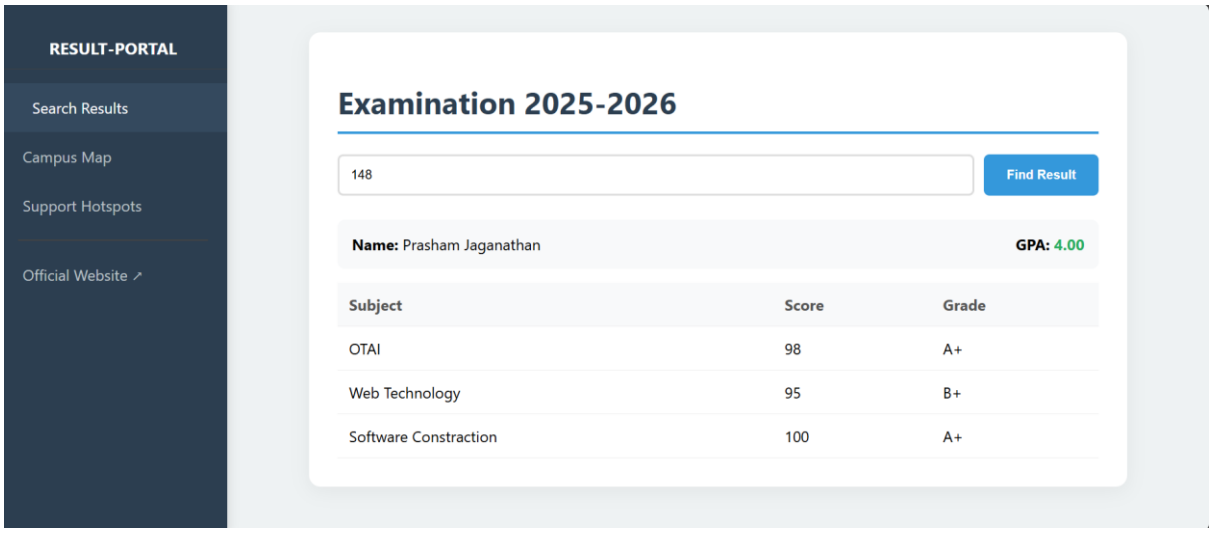
```
            </iframe>
        </div>
    </section>

    <section id="support-view" style="display:none;">
        <h1 style="color: var(--sidebar-color);">Support Hotspots</h1>
        <p>Click on the specific zone for help:</p>




        <div class="map-box">
            <img src="rec_logo.png"
                usemap="#helpmap" style="width: 100%;">


            <map name="helpmap">
                <area shape="rect" coords="0,0,400,200" alt="Academic" href="javascript:void(0)" onclick="alert('Navigating to Academic Support...')">
                <area shape="rect" coords="401,0,800,200" alt="Tech" href="javascript:void(0)" onclick="alert('Opening Tech Support Chat...')">
            </map>
        </div>
        <p style="font-size: 0.85rem; color: #666; margin-top: 10px;">
```

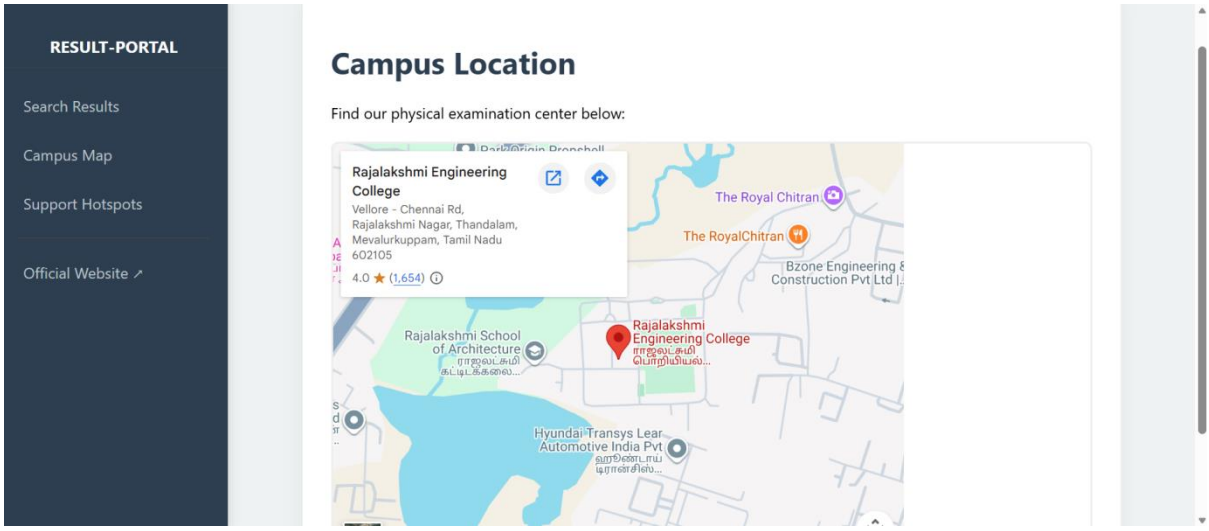
```
                * The Left Side (Zone A) and Right Side (Zone B) are interactive hotspots.
            </p>
        </section>


    </div>
  </main>


  <script>
    // MOCK DATABASE
    const database = {
        "149": { name: "Naresh kumar", gpa: "3.85", scores: [{ sub: "OTA", mark: 88, grade: "A" }, { sub: "Web Technology", mark: 96, grade: "B+" }, { sub: "Software Constraction", mark: 92, grade: "A+" }] },
        "147": { name: "Sri sidtharthan", gpa: "3.60", scores: [{ sub: "OTAI", mark: 75, grade: "B+" },{ sub: "Web Technology", mark: 93, grade: "B+" }, { sub: "Software Constraction", mark: 89, grade: "A" }] },
        "148": { name: "Prasham Jaganathan", gpa: "4.00", scores: [{ sub: "OTAI", mark: 98, grade: "A+" }, { sub: "Web Technology", mark: 95, grade: "B+" }, { sub: "Software Constraction", mark: 100, grade: "A+" }] }
    };


    // VIEW TOGGLE LOGIC
    function toggleView(view) {
        document.getElementById('search-view').style.display = (view === 'search') ? 'block' : 'none';
        document.getElementById('campus-view').style.display = (view === 'campus') ? 'block' : 'none';
        document.getElementById('support-view').style.display = (view === 'support') ? 'block' : 'none';
    }
```

```
    // FIND RESULT LOGIC
    function findResult() {
        const input = document.getElementById('rollNo').value.trim();
        const card = document.getElementById('resultCard');
        const error = document.getElementById('errorMsg');
        const tableBody = document.getElementById('resultTableBody');

        error.innerText = "";
        card.style.display = "none";
        tableBody.innerHTML = "";

        if (database[input]) {
            const student = database[input];
            document.getElementById('sName').innerText = student.name;
            document.getElementById('sGPA').innerText = student.gpa;

            student.scores.forEach(item => {
                tableBody.innerHTML += `<tr><td>${item.sub}</td><td>${item.mark}</td><td>${item.grade}</td></tr>`;
            });
            card.style.display = "block";
        } else {
            error.innerText = "Invalid Roll Number. Please use 101, 102, or 103.";
        }
    }
  </script>
</body>
</html>
```

OUTPUT:


RESULT-PORTAL
Search Results
Campus Map
Support Hotspots
Official Website ↗
Examination 2025-2026
Enter Roll Number (e.g. 101, 102, 103)
Find Result


RESULT-PORTAL
Search Results
Campus Map
Support Hotspots
Official Website ↗
Examination 2025-2026
148
Find Result
Name: Prasham Jaganathan
GPA: 4.00
Subject
Score
Grade
OTAI
98
A+
Web Technology
95
B+
Software Constraction
100
A+


RESULT-PORTAL
Search Results
Campus Map
Support Hotspots
Official Website ↗
Campus Location
Find our physical examination center below:
Rajalakshmi Engineering College
Vellore - Chennai Rd, Rajalakshmi Nagar, Thandalam, Mevalurkuppam, Tamil Nadu 602105
4.0 (1,654)
The Royal Chitran
The RoyalChitran
Bzone Engineering & Construction Pvt Ltd
Rajalakshmi School of Architecture
Rajalakshmi Engineering College
Hyundai Transys Lear Automotive India Pvt

Result:

Html Page is created with embedded Bangalore map along with hot spot, frames ,map & Links.